CONTACT OR CONTOUR? SUPERVISING A BOUNDARY MATTERS MORE THAN WHICH BOUNDARY FOR SEPARATING TOUCHING NUCLEI

Varun Kumar Dasoju[1], *Qingsu Cheng*[2], *Zeyun Yu*[1]

University of Wisconsin–Milwaukee, Milwaukee, WI, USA
[1]Department of Computer Science; [2]Department of Biomedical Engineering
vdasoju@uwm.edu, chengq@uwm.edu, yuz@uwm.edu

ABSTRACT

Supervising the borders where nuclei touch, as in the winning 2018 Data Science Bowl solution, is a popular way to keep touching nuclei from merging, but its effect is rarely separated from the design choices that change with it. We measure it in matched ten-seed experiments with image-paired confidence intervals on BBBC039, two NucleusSegData test sets (Huh7, HepG2), and 53 external DSB2018 images, against full-contour, semantic-watershed, and scratch-trained Cellpose baselines. Adding a boundary head, with marker-based decoding, to the watershed baseline raised aggregated Jaccard index (AJI) by 0.027–0.041 on BBBC039 and 0.035–0.036 externally, and object F1, the pre-specified endpoint, by 0.023–0.028 and 0.028–0.040. Which boundary was supervised mattered less: on BBBC039 a contact-only target beat the full-contour target by 0.014 AJI (46 of 50 images), robust to seed and image resampling and to test-time augmentation, with fewer merged touching nuclei (21% vs. 27%), but by only 0.005 in object F1. Elsewhere the contact advantage was small or conditional: externally the targets tied in AJI and contact gained 0.011 object F1; Huh7 excluded a gain of the BBBC039 size; HepG2 gained only after re-tuned post-processing; and a contact head lowered Cellpose's AJI. Scratch-trained Cellpose, which has no boundary head, had the best panoptic quality on every test set. For foreground-based networks, supervising a boundary is worth doing; which boundary is a secondary choice to confirm on held-out data.

Index Terms— nucleus instance segmentation, touching nuclei, boundary supervision, Cellpose, empirical comparison

1. INTRODUCTION

Nuclear instance masks support cell counting, morphometry, and spatial analysis. A segmentation that covers two touching nuclei with one object barely changes foreground Dice yet corrupts counts and per-cell measurements; on BBBC039, 29% of nuclei touch another. Instance methods separate nuclei with contour classes [1, 2], distance or shape representations [3, 4, 5], gradient flows [6], or multi-stage attention [7]. A simpler, widely used heuristic is to supervise where nuclei touch: U-Net up-weighted the gaps between touching cells [8], a multiclass loss added a touching-region class [9], and the winning 2018 Data Science Bowl solution predicted touching borders together with masks [10, 11].

New methods typically change backbone, decoder, targets, and training recipe together, and single-run comparisons can mistake seed variance for method effects [12]. We ask how much contact targets help relative to the choices that change with them, separating matched from composite contrasts. Our contributions are: (1) a ranking of design effects (Fig. 3): where both were measured, adding a boundary head raised AJI by 0.027–0.041 and object F1 by 0.023–0.040, while the boundary target changed AJI by at most 0.014 on any of four test sets, and not consistently; (2) a multi-seed, image-paired test of contact versus full-contour targets with identical network, loss, selection, and decoder, robust in AJI to seed and image resampling on BBBC039; and (3) an error analysis of where contact targets help (fewer merged touching nuclei on BBBC039) and where they do not (Huh7, inside Cellpose).

2. METHODS

Boundary targets. Let L be the reference instance map (0: background), $N_k(x)$ the k-neighbourhood of pixel x, and $\mathcal{D}$ dilation by a radius-1 disk. The two targets differ only in whether the neighbour that defines a boundary may be background:

$$C = \{x : L(x) > 0,\ \exists x' \in N_8(x) : L(x') \notin \{0, L(x)\}\},$$
$$B = \{x : L(x) > 0,\ \exists x' \in N_4(x) : L(x') \neq L(x)\}, \tag{1}$$

with contact target $c = \mathcal{D}(C) \cap \{L > 0\}$ and full-contour target $b = \mathcal{D}(B) \cap \{L > 0\}$ [1, 2]. Thus c keeps only borders between nuclei, the cue that decides separation, whereas b also marks nucleus–background edges that the foreground head already resolves; on BBBC039, c covers 0.34% of pixels and b 4.9%. Targets are built at native resolution and re-

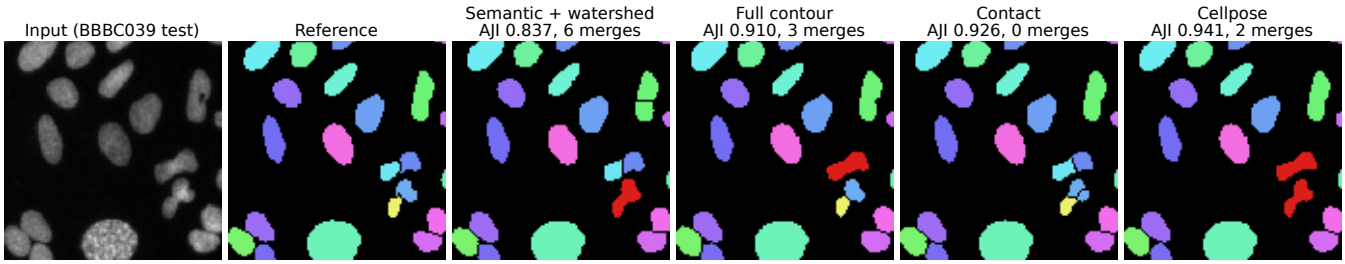


Fig. 1. BBBC039 test crop chosen by a fixed rule (contact model's upper-median seed by object F1, its median image, the window with most contacts; same seed for all models). Objects take the colour of the reference nucleus they mostly cover; red: merging objects; grey: unmatched. Titles: whole-image AJI, merging objects.

sized (nearest neighbour) with the image; the foreground target $y_f = \{L > 0\}$ is left intact.

Network and loss. A UNet++ [13] with an ImageNet-pretrained EfficientNet-B7 encoder [14] (decoder widths 256–16) ends in one two-channel layer whose sigmoids give foreground p_f and boundary p_c, trained on c or, for the control, on b. The loss is

$$\begin{aligned}\mathcal{L} = \tfrac{1}{2}[\mathcal{L}_D(p_f, y_f) + \mathcal{L}_B(p_f, y_f)] \\ + \tfrac{1}{4}[\mathcal{L}_D(p_c, c) + \mathcal{L}_{B,w}(p_c, c)],\end{aligned} \tag{2}$$

with soft Dice $\mathcal{L}_D$ and binary cross-entropy $\mathcal{L}_B$; $\mathcal{L}_{B,w}$ weights positive pixels by $w = \min(50, \max(1, (1-r)/r))$ for batch target ratio r, which sits at the cap for c and near 20 for b, so the target contrast includes this weighting difference.

Decoding. A boundary prediction may split a foreground component but never joins or deletes one: (i) $F = \{p_f > .5\}$ after filling holes and removing objects smaller than s pixels ($s = 16$; NucleusSegData: chosen on training images); (ii) markers are the 4-connected components of $F \setminus \{p_c > \tau\}$ with at least s pixels, $\tau \in \{.3, \ldots, .7\}$ chosen on validation; (iii) within each 4-connected component of F, every pixel takes its Euclidean-nearest marker in that component (component-preserving), or the component is flooded from its markers by watershed on p_c; (iv) a component without markers stays one object. Given the foreground, separation errors therefore arise only from missed boundaries (merges) or spurious ones (splits). An earlier global nearest-marker decoder, found faulty after test scoring, handed marker-less components to neighbours and more than doubled the contact–contour AJI gap (.870 vs. .838); no reported result uses it.

Matched controls and baselines. With identical network, loss, selection, and decoder, we compare c with b and with a *carved* target that, like one variant of the 2018 winning solution, removes contact pixels from y_f (decoded with $F = \{p_f > .5\} \cup \{p_c > \tau\}$). A one-channel *semantic* network is decoded by connected components (CC) or distance-transform watershed (h-maxima on a smoothed distance map, h, σ chosen on validation); its checkpoints are selected by validation object F1 after watershed, as for the two-head models. Cellpose [6] (v3.1.1) is trained from scratch on the same split (its released models saw Data Science Bowl images, which share BBBC039's source) with its default recipe (500 epochs; inference diameter: median square-root nucleus area, 25/32 pixels; Cellpose's convention, 28/36, changed AJI by $\leq$0.004); flow and probability thresholds are chosen on validation. We add contact supervision to it as a fourth output trained with the contact loss above (decoded normally or by splitting objects along predicted contacts), and as weights $1 + Wc$ on Cellpose's flow and probability losses, screened on validation ($W \in \{3, 10\}$) by a pre-set rule (object F1 $+0.002$, no seed declining).

Error taxonomy. A reference nucleus is *merged* if its majority lies in a predicted object that also holds another nucleus's majority, *split* if it holds the majority of two or more objects, and *unmatched* without an IoU$>$0.5 match; strata (touching another nucleus; small, $<$128 native pixels) were fixed before analysis. Rates pool nuclei per seed, then average over seeds.

3. EXPERIMENTAL SETUP

Data. BBBC039v1 has 200 520×696 Hoechst images of U2OS cells [15, 11]; we use the official 100/50/50 split. Expanded NucleusSegData [7, 16] has 25 training, 11 Huh7, and 25 HepG2 test images. For zero-shot testing we use the 53 grey-scale fluorescence images of the 2018 Data Science Bowl stage-1 test set [11], none duplicating a BBBC039 image (template-matching correlation $\leq$0.58).

Protocol. Base models resize percentile-normalized images to 512^2 and train for 20 epochs (AdamW, learning rate 3×10^{-4}, cosine decay, flips and rotations); checkpoints and τ are chosen on the 50 validation images by object F1 before test scoring. Base two-head rows re-decode these locked choices with the component-preserving decoder; TTA rows reselect decoder and τ on validation from one grid shared by both targets; all re-decodings are reported. A final setting trains at native resolution (random 512^2 crops, 40 epochs) and averages eight dihedral test-time augmentations (TTA); a

Table 1. BBBC039 official test set, mean (seed SD) over ten seeds unless marked ([§]five seeds). Two-head rows use the component-preserving decoder (final setting: it or watershed on p_c, chosen on validation); [¶]200 epochs, intensity jitter, chosen on validation. Literature: published values, same split; [‡]scored on expanded-interior targets.

Model	DICE1	AJI	PQ	Obj.-F1
Foreground-based, base protocol (512^2)				
Semantic + CC	.9653(.0013)	.6779(.0133)	.7406(.0080)	.7520(.0084)
Semantic + watershed	.9653(.0013)	.8303(.0050)	.8371(.0029)	.8583(.0031)
Full-contour target	.9637(.0016)	.8568(.0042)	.8541(.0044)	.8817(.0044)
Carved foreground	.9650(.0025)	.8692(.0036)	.8619(.0039)	.8865(.0033)
Contact target	.9660(.0023)	.8712(.0057)	.8610(.0047)	.8865(.0040)
Final setting (TTA, validation-selected decoder)				
Full contour, 512^2	.9676(.0006)	.8682(.0033)	.8664(.0025)	.8934(.0027)
Contact, 512^2	.9692(.0010)	.8792(.0051)	.8713(.0029)	.8959(.0030)
Contact, native resolution	.9715(.0008)	.8895(.0040)	.8807(.0028)	.9049(.0023)
+ searched recipe[¶]	**.9721**(.0003)	.8965(.0023)	.8835(.0016)	.9069(.0013)
Flow-based (Cellpose 3, from scratch)				
Cellpose	.9702(.0016)	**.9053**(.0035)	**.8884**(.0036)	.9119(.0031)
+ contact head[§]	.9700(.0004)	.8975(.0013)	.8872(.0009)	**.9121**(.0011)
ASW-Net[‡] [19]	.9645	.9020	.8665	–
BISSG [20]	.9482	.8680	.8629	–
Caicedo U-Net [2]	–	–	–	.8980

validation-only search (six recipes, two seeds) then chose 200 epochs with intensity jitter. NucleusSegData has no validation set: models train for 60 epochs (Cellpose: 500), keep the final epoch, and choose post-processing on training images.

Endpoints and statistics. Foreground Dice (DICE1), AJI [17], panoptic quality (PQ) [18], and object F1 over IoU 0.50:0.05:0.90 [2], as image means; NucleusSegData uses the Bilkent nucleus F1 [16] (one-to-one matches covering half of each other, pooled per subset). The locked protocols designated object F1 as primary; AJI, a main endpoint of prior BBBC039 work [19, 20], was emphasized after the locked runs and is reported alongside it. Contact vs. contour (base protocol, decoder corrected after test scoring) is the confirmatory contrast; all other intervals, including boundary-head effects, are exploratory and unadjusted. Count error is the mean absolute per-image count difference. Seeds are averaged within each test image; 95% percentile CIs use 10,000 image-paired bootstrap resamples, conditional on the trained models; a two-way bootstrap that also resamples each arm's seeds checks retraining variability.

4. RESULTS

4.1. Supervising a boundary mattered more than which one

Supervising contacts instead of contours added 0.014 AJI ($[.011, .018]$; 46/50 images) and object F1 $+0.005$ ($[.002, .008]$); with TTA (same decoders and selection for both), 0.011 ($[.008, .014]$; 45/50; object F1 $+0.003$, $[-.0001, .005]$). Resampling seeds as well as images kept

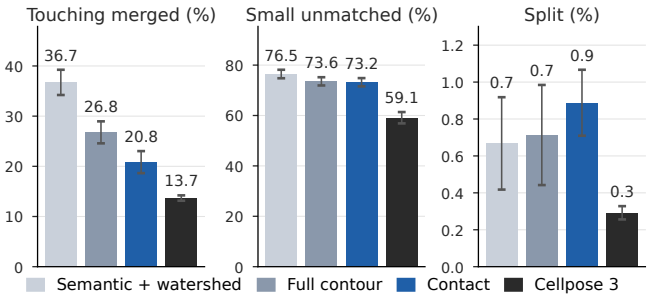


Fig. 2. Error profile on BBBC039 test (base protocol; mean and seed SD over ten seeds) as a percentage of the 1,650 touching, 509 small (<128 pixels), and 5,720 reference nuclei.

both AJI intervals above zero ($[.009, .020]$, $[.006, .016]$), but not the object F1 interval with TTA ($[-.001, .006]$; base $[.0002, .010]$). Adding a boundary head[†] to the watershed baseline mattered more (exploratory; the decoder changes with it): AJI rose by 0.041 ($[.032, .051]$; 48/50) with contact and 0.027 ($[.019, .035]$) with contour, 0.012 ($[.004, .021]$) more than the target effect, and object F1 by 0.028 ($[.022, .035]$) and 0.023 ($[.017, .030]$), 0.019 ($[.012, .026]$) more. On identical semantic predictions, a tuned watershed instead of connected components (a no-separation floor) raised AJI by 0.152 ($[.126, .178]$). Correcting the faulty decoder moved AJI by 0.018 for contour networks ($[.015, .022]$; 49/50) but 0.001 for contact networks. Carving contacts out of the foreground changed AJI by 0.002 ($[.0004, .0036]$), and a validation-searched recipe raised it by 0.007 ($[.004, .010]$). Cellpose exceeded our best contact model by 0.009 ($[.004, .013]$; seed and image resampling $[.004, .014]$).

4.2. On BBBC039, contact targets merged fewer nuclei

Contact supervision was accompanied by (Fig. 2) fewer merges of touching nuclei, from 36.7% (watershed) and 26.8% (full contour) to 20.8% (contact minus contour -5.9 points, $[-7.7, -4.4]$, seed and image resampling $[-8.6, -3.5]$; 44/50 images), and of all nuclei, from 10.8% and 8.2% to 6.3%, while splits stayed below 1% (0.67%, 0.71%, 0.89%), although splits of touching nuclei rose from 1.3% (contour) to 2.3%. All three models undercounted, and the per-image count error fell from 10.1 (watershed) and 7.2 (contour) to 6.6 nuclei. Unmatched small nuclei, by contrast, were similar across foreground-based models (73–77%); training at native resolution lowered this to 64% for the contact model (no TTA), so resizing explains only part of it.

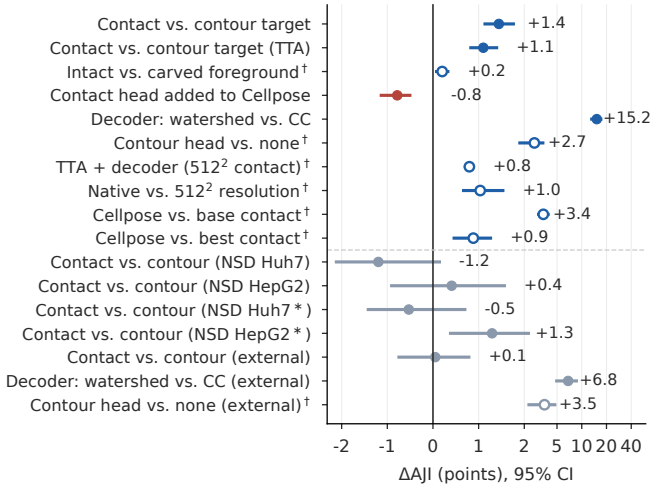


Fig. 3. Effect of each design choice on AJI (points, AJI×100; mean of image-paired differences, 95% bootstrap CI; symmetric-log axis). Rows above the dashed line use BBBC039 test; below it, NucleusSegData (NSD; *widened grid and TTA for both targets) and the external DSB2018 images. Open markers (†): composite comparisons that also change decoder, reconstruction, recipe, or resolution.

Table 2. Expanded NucleusSegData, Bilkent nucleus F1; ours: mean (seed SD) over ten seeds, component-preserving decoder; AttentionBoost: five published runs [7]. *Widened training-selected grid and TTA. §Five seeds.

Method	Huh7	HepG2
Multi-task [7]	.934(.009)	.869(.005)
AttentionBoost [7]	.954(.005)	.919(.008)
Full contour	.9541(.0062)	.9162(.0079)
Contact	.9533(.0057)	.9170(.0080)
Full contour*	.9609(.0039)	.9240(.0029)
Contact*	.9575(.0051)	.9286(.0069)
Cellpose 3, scratch§	**.9715**(.0038)	**.9426**(.0038)

4.3. A contact head did not help Cellpose

Cellpose merged 13.7% of touching nuclei, split 0.29% of all nuclei, and left fewer small nuclei unmatched (59%; Fig. 2). In the tested recipe, the auxiliary contact head (five seeds) lowered AJI relative to the ten baseline seeds by 0.008 ($[.005, .012]$; worse on 39/50 images) without materially changing PQ or object F1 (-0.001, $+0.000$)(equal seeds, resampled: -0.007, $[-.011, -.002]$); validation never selected contact-based splitting in any seed. The contact-weighted loss failed its validation screen and was not tested.

4.4. Second benchmark and external images

On expanded NucleusSegData (Table 2), the contact model matched the published multi-stage AttentionBoost in nucleus F1 and was higher with a widened training-selected post-processing grid and TTA. In the base setting the targets did not differ detectably (Huh7 AJI -0.012, $[-.024, .002]$, 2/11 images; HepG2 $+0.004$, $[-.009, .016]$), with nucleus F1 differences below 0.001; the Huh7 interval excludes the BBBC039 gain of 0.014. Re-decoding both targets with the widened grid and TTA (Table 2, *) kept Huh7's direction (-0.005, $[-.015, .007]$) but favoured contact on HepG2 ($+0.013$, $[.004, .023]$; 16/25; nucleus F1 $+0.005$). Scratch-trained Cellpose reached .971 and .943, 1.1 and 1.4 points above the best foreground-based model, and the best PQ (.870 and .795 vs. at most .859 and .785). Applied unchanged to the 53 external images (Table 3), a boundary head again improved on the semantic watershed in AJI (contour $+0.035$, $[.022, .049]$; contact $+0.036$) and object F1 ($+0.028$, $+0.040$). The targets tied in AJI ($+0.001$, $[-.008, .008]$) and PQ ($+0.008$, $[-.0001, .016]$); contact had higher object F1 ($+0.011$, $[.002, .021]$; 35/53) but merged more touching nuclei (30.9% vs. 27.6%; $+3.3$ points, $[1.3, 4.8]$), a difference not robust to seed resampling ($[-0.4, 7.3]$). Cellpose exceeded the contact model in PQ ($+0.044$, $[.016, .072]$) but not detectably in AJI ($+0.012$, $[-.016, .039]$); its count error was larger by 4.4 nuclei per image ($[2.4, 6.9]$), yet it left far fewer predicted objects unmatched (pooled 12% vs. 24%), so the foreground models' smaller count error partly reflected unmatched predictions offsetting missed nuclei.

Table 3. Zero-shot results on 53 external DSB2018 fluorescence images (BBBC039-trained base models, ten seeds; count error in nuclei per image).

Model	DICE1	AJI	PQ	Obj.-F1	Count err.
Semantic + watershed	.803	.585	.555	.493	12.2
Full-contour target	**.812**	.620	.581	.521	**9.1**
Contact target	.811	.621	.589	.532	9.3
Cellpose	.796	**.633**	**.633**	**.584**	13.7

5. DISCUSSION AND CONCLUSION

Where both were measured, supervising a boundary helped more than the choice of boundary, in AJI and in the prespecified object F1. Contact beat contour robustly in AJI on BBBC039, on HepG2 only after re-tuned post-processing, and externally only in object F1; it did not help on Huh7 or inside Cellpose. Cellpose, without a boundary head, was the strongest pipeline (best PQ on every test set), so boundary supervision is a lever for foreground-based networks, not a requirement. In practice: add a boundary head, tune the decoder (never reassign marker-less components), compare with a flow-based model, and confirm the target choice on held-out data. Limitations: fluorescence images only, one annotation each; the targets differ in effective positive weight; the decoder fix followed test scoring; the native-resolution and searched settings lack a contour arm; one Cellpose recipe; no StarDist [4] or foundation-model baselines [21, 22].

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using open-access images of cultured cell lines (BBBC039 [15], NucleusSegData [7], DSB2018 [11]). It involved no human subjects or animals, and ethical approval was not required.

7. ACKNOWLEDGMENTS

This work was supported by U.S. Department of Energy grant DE-SC0025403. The authors declare no competing interests. OpenAI Codex and Anthropic Claude assisted with analysis scripts, statistics, figures, and language editing of all sections; the authors designed and verified the experiments and take responsibility for the manuscript. Code and per-image results: `https://github.com/vkd14/contact-or-contour`.

8. REFERENCES

[1] Hao Chen, Xiaojuan Qi, Lequan Yu, et al., "DCAN: Deep contour-aware networks for accurate gland segmentation," in *CVPR*, 2016, pp. 2487–2496.

[2] Juan C. Caicedo et al., "Evaluation of deep learning strategies for nucleus segmentation in fluorescence images," *Cytometry Part A*, vol. 95, no. 9, pp. 952–965, 2019.

[3] Peter Naylor, Marick Lae, Fabien Reyal, et al., "Segmentation of nuclei in histopathology images by deep regression of the distance map," *IEEE Transactions on Medical Imaging*, vol. 38, no. 2, pp. 448–459, 2019.

[4] Uwe Schmidt, Martin Weigert, Coleman Broaddus, et al., "Cell detection with star-convex polygons," in *MICCAI*, 2018, pp. 265–273.

[5] Simon Graham et al., "HoVer-Net: Simultaneous segmentation and classification of nuclei in multi-tissue histology images," *Medical Image Analysis*, vol. 58, pp. 101563, 2019.

[6] Carsen Stringer, Tim Wang, Michalis Michaelos, et al., "Cellpose: A generalist algorithm for cellular segmentation," *Nature Methods*, vol. 18, pp. 100–106, 2021.

[7] Gozde Nur Gunesli, Cenk Sokmensuer, and Cigdem Gunduz-Demir, "AttentionBoost: Learning what to attend for gland segmentation in histopathological images by boosting fully convolutional networks," *IEEE Transactions on Medical Imaging*, vol. 39, no. 12, pp. 4262–4273, 2020.

[8] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *MICCAI*, 2015, pp. 234–241.

[9] Fidel A. Guerrero-Peña, Pedro D. Marrero Fernandez, Tsang Ing Ren, Mary Yui, Ellen Rothenberg, and Alexandre Cunha, "Multiclass weighted loss for instance segmentation of cluttered cells," in *ICIP*, 2018, pp. 2451–2455.

[10] Victor Durnov, Selim Seferbekov, and Alexander Buslaev, "DSB2018 [ods.ai] topcoders 1st place solution," `https://github.com/selimsef/dsb2018_topcoders`, 2018.

[11] Juan C. Caicedo et al., "Nucleus segmentation across imaging experiments: The 2018 data science bowl," *Nature Methods*, vol. 16, pp. 1247–1253, 2019.

[12] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, et al., "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, 2021, vol. 3, pp. 747–769.

[13] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, et al., "UNet++: A nested U-Net architecture for medical image segmentation," in *DLMIA*, 2018, pp. 3–11.

[14] Mingxing Tan and Quoc V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *ICML*, 2019, pp. 6105–6114.

[15] Vebjorn Ljosa, Katherine L. Sokolnicki, and Anne E. Carpenter, "Annotated high-throughput microscopy image sets for validation," *Nature Methods*, vol. 9, no. 7, pp. 637, 2012.

[16] Can Fahrettin Koyuncu, Rengul Cetin-Atalay, and Cigdem Gunduz-Demir, "Object-oriented segmentation of cell nuclei in fluorescence microscopy images," *Cytometry Part A*, vol. 93, no. 10, pp. 1019–1028, 2018.

[17] Neeraj Kumar, Ruchika Verma, Sanuj Sharma, et al., "A dataset and a technique for generalized nuclear segmentation for computational pathology," *IEEE Transactions on Medical Imaging*, vol. 36, no. 7, pp. 1550–1560, 2017.

[18] Alexander Kirillov, Kaiming He, Ross Girshick, et al., "Panoptic segmentation," in *CVPR*, 2019, pp. 9404–9413.

[19] Weihao Pan, Zhe Liu, Weichen Song, et al., "An integrative segmentation framework for cell nucleus of fluorescence microscopy," *Genes*, vol. 13, no. 3, pp. 431, 2022.

[20] Xiaoyu Liu, Wei Huang, Yueyi Zhang, et al., "Biological instance segmentation with a superpixel-guided graph," in *IJCAI*, 2022, pp. 1209–1215.

[21] Anwai Archit et al., "Segment anything for microscopy," *Nature Methods*, vol. 22, pp. 579–591, 2025.

[22] Marius Pachitariu, Mike Rariden, and Carsen Stringer, "Cellpose-SAM: Superhuman generalization for cellular segmentation," *bioRxiv*, 2025.